1. Prove cash additivity

$$\rho(L+c) = \rho(L) + c$$

set

$$\rho(L+c) = \frac{1}{\gamma} \ln E[e^{\gamma(L+c)}]$$

propt. $a^{a+b} = e^{a}e^{b}$

$$\leq \frac{1}{\gamma} \ln E[e^{\gamma L} e^{\gamma c}]$$

$$= \frac{1}{\gamma} \ln \left(e^{\gamma c} E[e^{\gamma L}]\right)$$

propt: $\ln(ab) = \ln(a) + \ln(b)$

$$= \frac{1}{\gamma} \left(\gamma c + \ln E[e^{\gamma L}]\right)$$

$$= c + \frac{1}{\gamma} \ln E[e^{\gamma L}] = c + \rho(L)$$

Prove monotonicity

$$L_1 \leq L_2 \Rightarrow \rho(L_1) \leq \rho(L_2)$$

If $\gamma > 0$, exponential function strictly increase, then:

$$e^{\gamma L_1} \leq e^{\gamma L_2}$$

take expected value:

$$E[e^{\gamma L_1}] \leq E[e^{\gamma L_2}]$$

$$\ln E[e^{\gamma L_1}] \leq \ln E[e^{\gamma L_2}]$$

$\times \frac{1}{\gamma}$ $\times \frac{1}{\gamma}$

prove $\rho(L_1) \leq \rho(L_2)$

2.

$$\rho(\alpha L + (1-\alpha)L_2) \leq \alpha\rho(L_1) + (1-\alpha)\rho(L_2)$$

Set $0 < \alpha < 1$ for $\rho(\alpha L_1 + (1-\alpha)L_2)$

$$= \frac{1}{\gamma} \ln E[e^{\gamma(\alpha L_1 + (1-\alpha)L_2)}]$$

$$= \frac{1}{\gamma} \ln E[(e^{\gamma L_1})^{\alpha}(e^{\gamma L_2})^{1-\alpha}]$$

use Hölder's Inequality

$$E[|XY|] \le (E[|X|^p])^{1/p} (E[|Y|^q])^{1/q}$$

where: $\frac{1}{p} + \frac{1}{q} = 1$, choose $p = \frac{1}{\alpha}$, $q = \frac{1}{1-\alpha}$

set

$$X = (e^{\gamma L_1})^{\alpha}, \quad Y = (e^{\gamma L_2})^{1-\alpha}$$

$$E[XY] \le (E[e^{\gamma L_1}])^{\alpha} (E[e^{\gamma L_2}])^{1-\alpha}$$

$$\rho(\alpha L_1 + (1-\alpha)L_2) \le \frac{1}{\gamma} \ln [(E[e^{\gamma L_1}])^{\alpha} (E[e^{\gamma L_2}])^{1-\alpha}]$$

$$\Downarrow$$

$$= \frac{\alpha}{\gamma} \ln E[e^{\gamma L_1}] + \frac{1-\alpha}{\gamma} \ln E[e^{\gamma L_2}]$$

$$\Downarrow$$

$$\alpha \rho(L_1) + (1-\alpha)\rho(L_2)$$

2.

Get MGF: $E[e^{tL}] = e^{t^2/2}$

first calculate $\rho(L)$:

$$\rho(L) = \frac{1}{\gamma} \ln E[e^{\gamma L}]$$

$$\Downarrow$$

$$= \frac{1}{\gamma} \ln (e^{\gamma^2/2})$$

thus $\ln(e^{\gamma}) = \gamma$

$$\rho(L) = \frac{\gamma}{2}$$

Second cal $\rho(\lambda L)$

$$\rho(\lambda L) = \frac{1}{\gamma} \ln E[e^{\gamma \lambda L}]$$

$$= \frac{1}{\gamma} \ln (e^{\gamma^2 \lambda^2/2})$$

$$p(\lambda L) = \frac{r\lambda^2}{2}$$

third test positive Homogeneity

If coherent, fill: $\lambda p(L) = \frac{r\lambda}{2}$

but $p(\lambda L) = \frac{r\lambda^2}{2} \neq \frac{r\lambda}{2}$

take $\lambda = 2$

$$p(2L) = 2r$$

and $2p(L) = r$

so $p(2L) \neq 2p(L)$